

Siddharth Sule

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Final-year PhD student (completing in July 2026) specialising in Monte Carlo simulation and high-performance scientific computing, with experience in mathematical modelling, GPU acceleration, and research software development. Seeking research-focused roles in particle physics, HPC, and related quantitative fields.

Education

PhD in Particle Physics

Sep 2022 – Jul 2026 (expected)

The University of Manchester

Supervised by Professor Michael H. Seymour

- Combining Monte Carlo simulation techniques with object-oriented programming to model high-energy particle collisions at the Large Hadron Collider (CERN). Contributing to the development of specialised simulation software, known as *event generators*, which provide predictions for experimental data.
- Specialising in simulating the radiation produced by quarks and gluons, known as a *parton shower*.

MPhys (Hons) Physics, First Class

Sep 2018 – Jun 2022

The University of Manchester

- *Integrated BSc+MSc Course*. Covered advanced statistical physics, scientific computing, and mathematical modelling, alongside core physics topics. For the MPhys project, simulated the spread of infectious diseases on community network models using agent-based models and analysed the impact of different intervention strategies.

Work Experience and Projects

GAPS: a GPU-Amplified Parton Shower

Mar 2024 – Present

Head Developer and Author | *Publications below* | gitlab:siddharthsule/gaps

- CERN's computing demands will soon outgrow the available budget, with simulation among the most resource-intensive tasks. Investigated GPU-accelerated strategies for Monte Carlo event generation to address this.
- Redesigned the parton shower algorithm for parallelised simulation and developed a CUDA C++ event generator that, on a single NVIDIA V100 GPU, matches the runtime and energy consumption of a 96-core CPU.

Improving Parton Shower Physics in Herwig 7

Sep 2023 – Present

Junior Developer and Project Lead | *Publication below* | herwig.hepforge.org

- Serving as a junior developer for *Herwig*, an open-source event generator used worldwide in particle physics research for over 40 years.
- To meet the precision demands of future experiments, integrated the newly developed Pan-Global and FHP parton showers with the cluster hadronisation model in Herwig, producing more reliable hadron-level simulations.

Teaching and Assessment Experience

Fellow of the Higher Education Academy (FHEA)

Mar 2026

Undergraduate Lab Demonstration

Feb 2023 – Sep 2025

The University of Manchester

- Facilitated laboratory experiments for over one hundred undergraduate students, assessing understanding through viva-style interviews and marking lab reports with a detailed rubric to evaluate analysis, reasoning, and scientific writing.
- Analysed marks from over one thousand laboratory reports in Python using a linear mixed-effects model, demonstrating a statistically significant dependence on allocated marker and motivating improvements in marking consistency.

Herwig Event Generator Tutorials

MCnet Summer School

Jun 2024, 2025, 2026

Terascale Monte Carlo School

Feb 2024, Nov 2025

- Introduced the Herwig event generator to PhD students and postdoctoral researchers, guiding attendees through simulating particle-collision events, customising physics settings, and comparing results with Large Hadron Collider data.

Skills Overview

Technical Skills: Scientific research, Linux, software development (C++, CUDA C++, Python), GPU/HPC programming, data analysis and statistical modelling, Git, CMake

Professional Skills: Project leadership (GAPS, Herwig developments), teaching at international schools, scientific writing and peer-reviewed publication, cross-institution collaboration

Highlighted Publications

Note: Particle physics publications list author names in alphabetical order.

Studying the Infrared Behaviour of Improved Logarithmic Accuracy Parton Showers with Herwig

May 2026

J. Holguin, S. Plätzer, M. H. Seymour and S. Sule, [arXiv:2605.02622](#)

An NLO-Matched Initial and Final State Parton Shower on a GPU

May 2026

M. H. Seymour and S. Sule, [SciPost Physics Codebases 71](#)

An Algorithm to Parallelise Parton Showers on a GPU

Aug 2024

M. H. Seymour and S. Sule, [SciPost Physics Codebases 33](#)

For more publications, visit <https://inspirehep.net/authors/2768399>